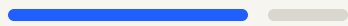


# YOLOE

## Real-Time Seeing Anything



DSC3032 Deep Learning 1 Final Project

### Group 7

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# OUTLINE

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challenges + limits

# PROBLEM

## Traditional YOLO is fast, but inflexible

- Trained on predefined categories (e.g., COCO's 80 classes)
- Cannot detect objects outside the training vocabulary
- Fails in open-world scenarios: novel objects, specialized domains, user-defined queries

## Open-set methods exist, but each has a cost

- Text Prompted : GLIPHeavy cross-modal fusion; slow with large vocabularies
- Visual Prompted : Transformer-heavy; poor edge deployability
- Prompt-free : Requires large language models; high memory and latency

**The gap : No single model supports *all three* prompt types with high efficiency and accuracy.**

# YOLOE: Performance Overview

## Text Prompt RepRTA

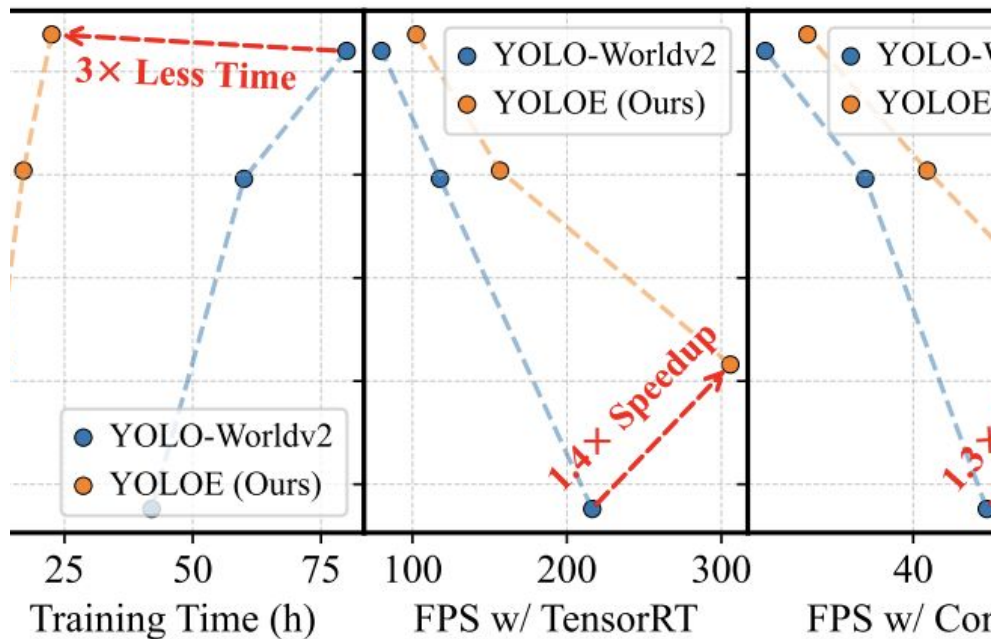
Input category names  
(ex. "dog, cat, person")

## Visual Prompt SAVPE

Point on a target  
(ex. bounding box)

## Prompt-free LRPC

No input needed



## How it compares to YOLO-World

**3x**

less training time vs  
YOLO-Worldv2-S

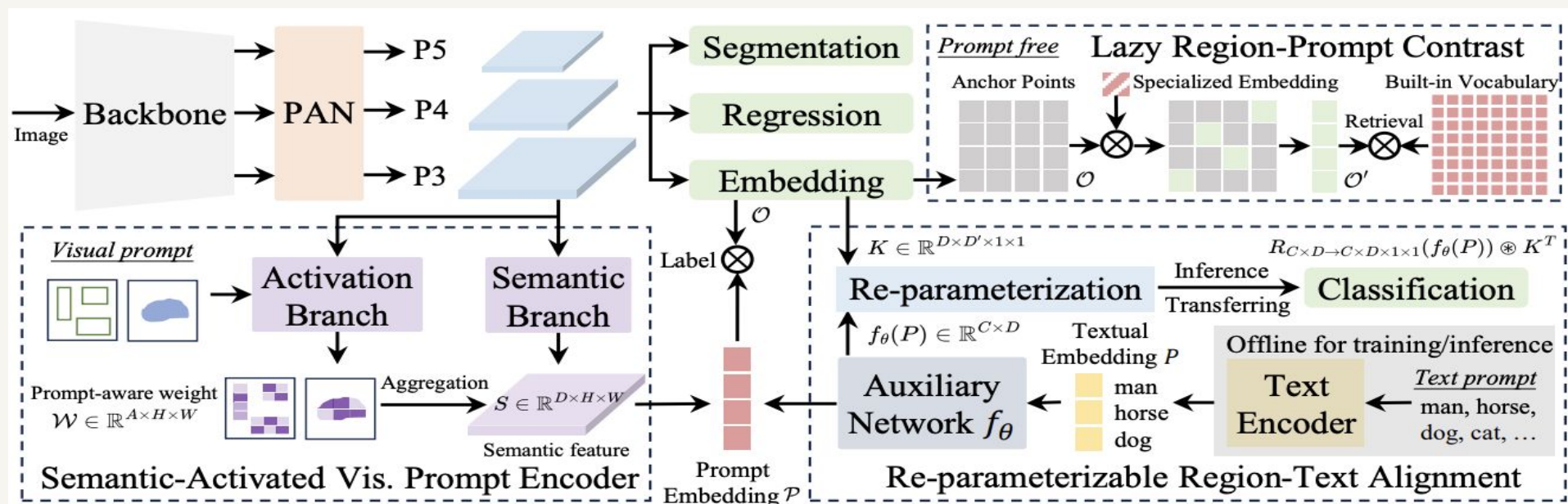
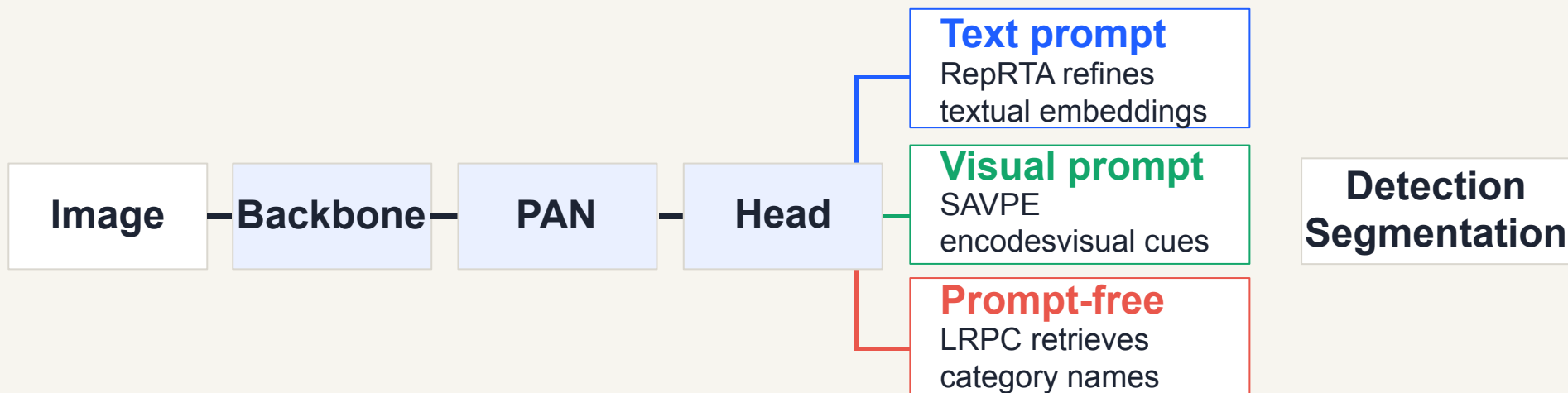
**1.4x**

faster TensorRT inference

**+3.5 AP**

on LVIS minival  
(zero-shot, S-scale)

# Architecture



# RepRTA: Text Prompt

## Problem

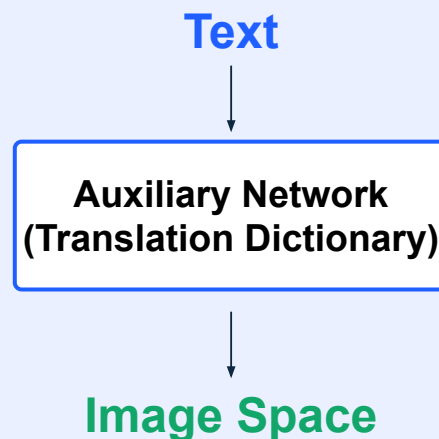
- Image features come from pixels
- Text features come from language
- Raw dot product may not align
- Existing methods mix the image and text features in real time for every input image

**Cross-modal operation =  
Category Counts x Image Locations**



## Idea

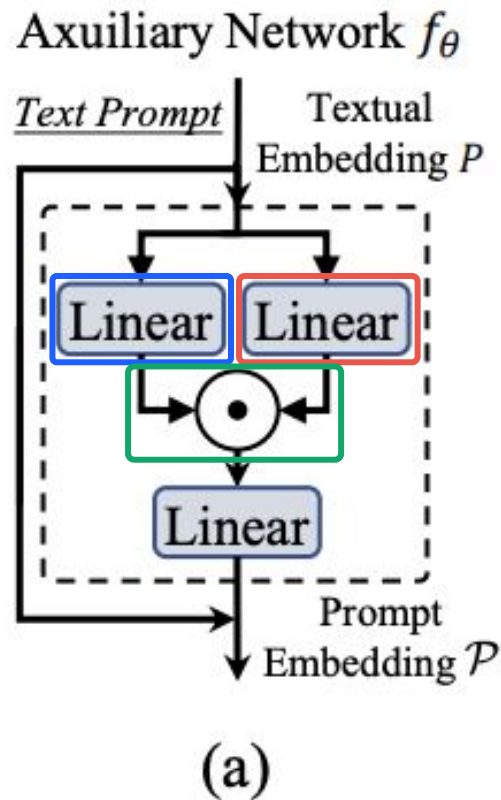
- Learn the mapping “rule” — apply to any text
- Reuse it for unseen words



# RepRTA: Auxiliary Network

## <Transform path>

Encodes text features into a common representation space.



## <Gate path>

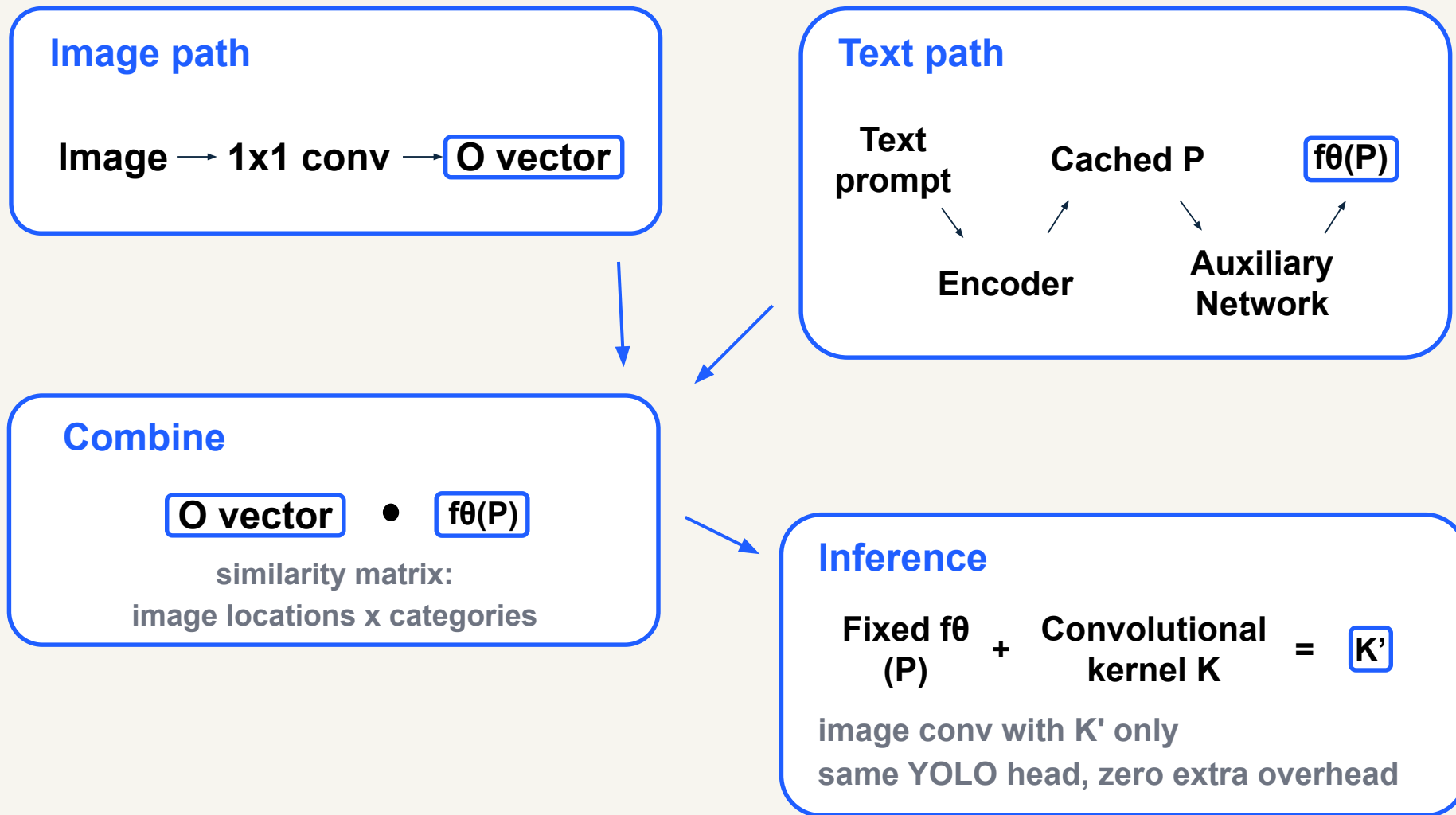
Decides how much each dimension passes through.

*"dog, cat" → animal dims*  
*"bus, truck" → car dims*

## <Element-wise $\times$ >

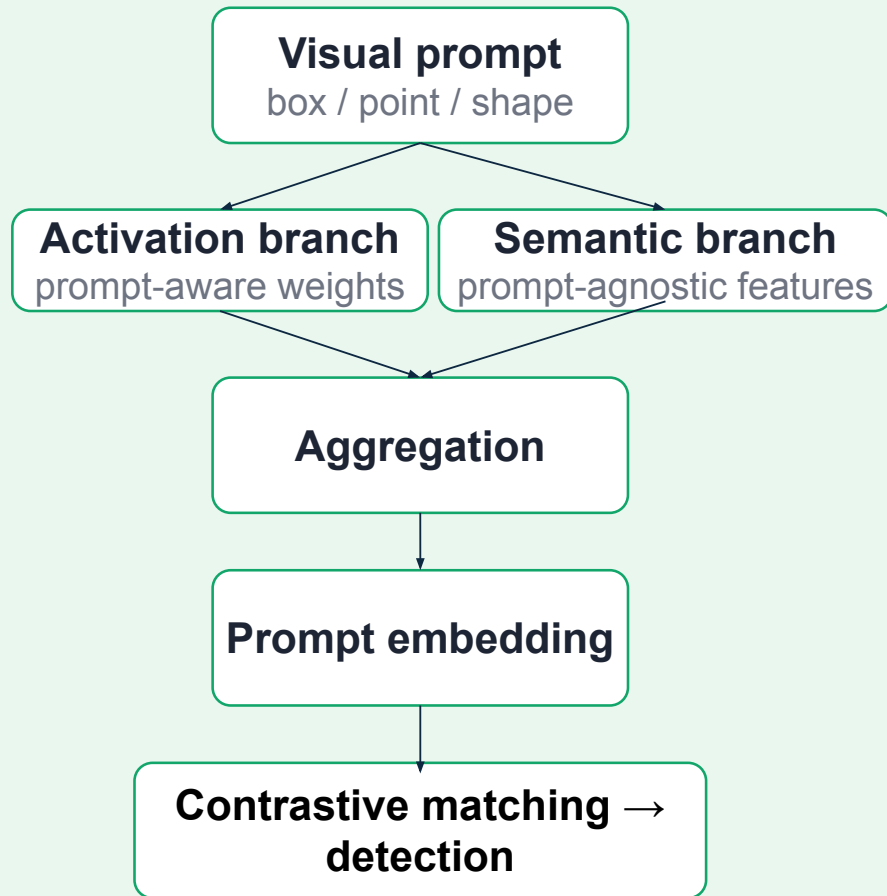
Keep only what matters for the current prompt.

# RepRTA: Flow Overview

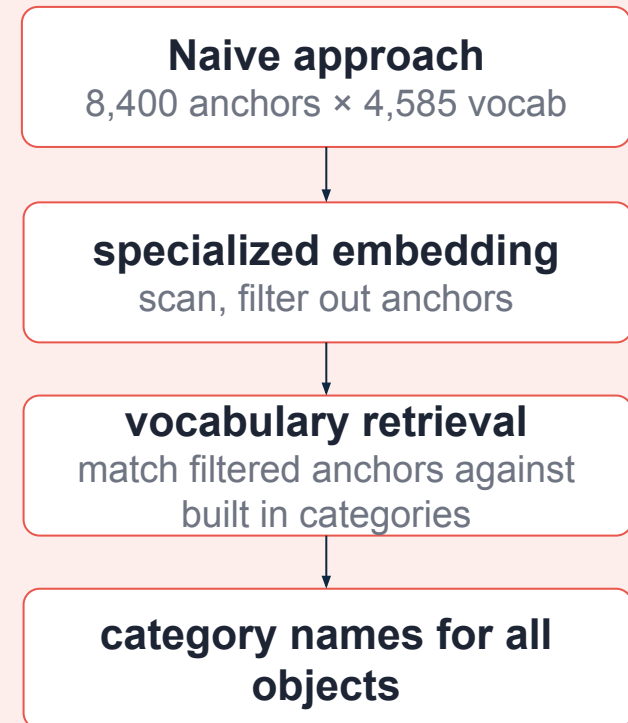


# SAVPE and LRPC

## SAVPE: visual prompt



## LRPC: prompt-free



# EXPERIMENTAL DESIGN

YOLOE's core claim is **open-vocabulary generalization**.

We reproduced two experiments:

one to **validate** the claim, one to **demonstrate** training.

## Zero-shot evaluation

- Compares pretrained model performance on **LVIS minival**.
- No training required
- Directly tests the paper's **core open-vocabulary claim**.

## COCO fine-tuning

- Fine-tunes on a **COCO subset** and evaluates standard AP.
- Added to include a **training experiment** in our reproduction.

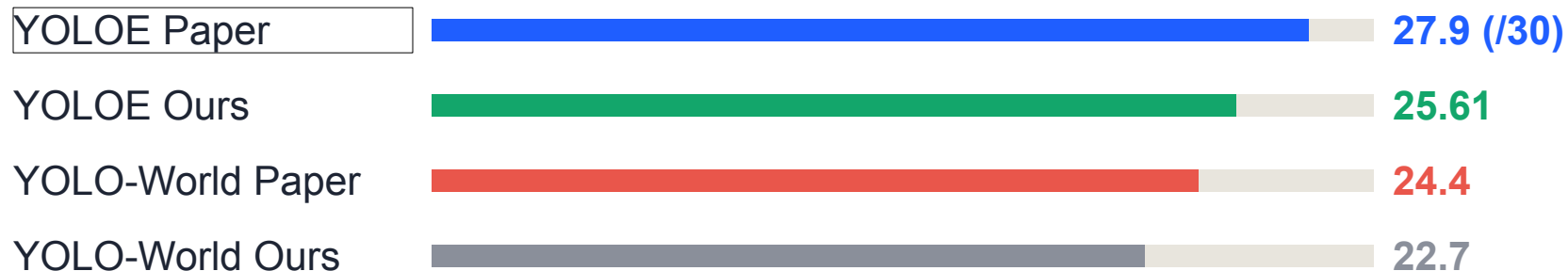
# IMPLEMENTATION DETAILS

| Item                           | Paper                                   | Ours                                           |
|--------------------------------|-----------------------------------------|------------------------------------------------|
| Model                          | YOLOE-v8-S / YOLO-World-v2-S            | YOLOE-v8-S / YOLO-World-v2-S                   |
| Eval dataset<br>(Table 1)      | LVIS minival (1,203 cats)               | LVIS minival (1,203 cats)                      |
| Fine-tune dataset<br>(Table 4) | Full COCO                               | Full COCO                                      |
| LP / Full FT<br>epochs         | 10 / 160                                | <b>10 / ~104</b>                               |
| Batch size                     | 128                                     | <b>64 (LP), 16 (FT), nbs=128</b>               |
| GPU                            | 8x RTX4090                              | 1x RTX 4090 (SKKU GPU)                         |
| FPS / Eval Metric              | TensorRT on T4<br>/ Fixed & Standard AP | RTX 4090 with PyTorch<br>/ Fixed & Standard AP |

**Resource constraint:** Due to compute constraints (1x vs 8x GPU, reduced batch and epochs), absolute AP gaps are expected. Directional trends are the focus.

# RESULTS (1) ZERO-SHOT EVALUATION

| Model                 | AP    | APr   | APc   | APf   | FPS   |
|-----------------------|-------|-------|-------|-------|-------|
| YOLOE-v8-S Paper      | 27.9  | 22.3  | 27.8  | 29.0  | 305.8 |
| YOLOE-v8-S Ours       | 25.61 | 18.95 | 25.79 | 26.65 | 86.6  |
| YOLO-World-v2-S Paper | 24.4  | 17.1  | 22.5  | 27.3  | 216.4 |
| YOLO-World-v2-S Ours  | 22.70 | 16.30 | 20.80 | 25.50 | 73.7  |

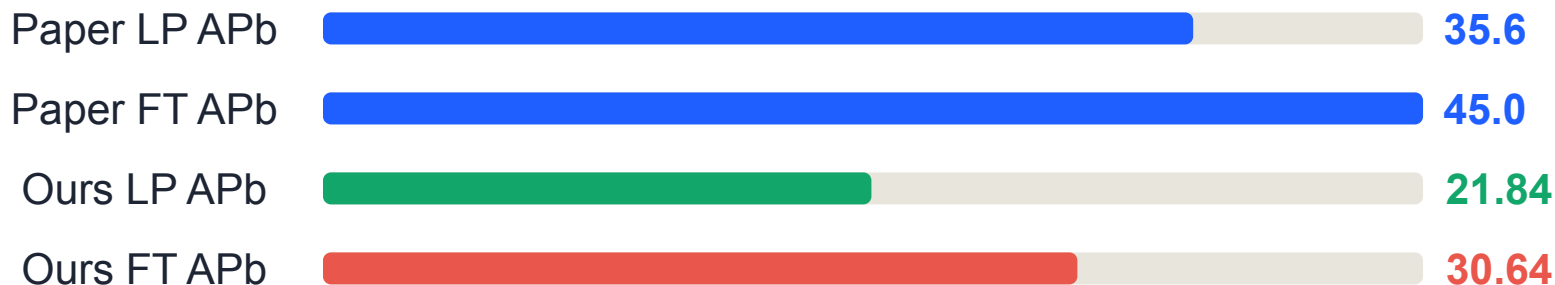


Likely gap sources: inference threshold details, TensorRT vs PyTorch, and other small pipeline settings.

# RESULTS (2) COCO FINE-TUNING

## Updated: Extended Training Epochs for Ours FT Training (~104 -> 160)

| Setting  | Dataset   | Epochs | Batch | APb   | APb50 | APm   | APm50 |
|----------|-----------|--------|-------|-------|-------|-------|-------|
| Paper LP | Full COCO | 10     | 128   | 35.6  | 51.5  | 30.3  | 48.2  |
| Paper FT | Full COCO | 160    | 128   | 45.0  | 61.6  | 36.7  | 58.3  |
| Ours LP  | Full COCO | 10     | 64    | 21.84 | 30.88 | 19.28 | 30.25 |
| Ours FT  | Full COCO | 160    | 16    | 30.64 | 42.90 | 26.65 | 41.89 |



**FT > LP pattern reproduced.** Absolute gap explained by batch size.  
When examining epoch-wise validation performance,  
**overfitting** was observed starting from epoch 60.

# DEMO - Table 4

## Linear Probing training log and final evaluation result

```
=====
LP Full COCO (batch=64, 10 epochs) - Training Progress
=====
Epoch 1: APb=26.53, APm=21.93
Epoch 2: APb=26.53, APm=21.93
Epoch 3: APb=26.53, APm=21.93
Epoch 4: APb=26.53, APm=21.93
Epoch 5: APb=26.53, APm=21.93
Epoch 6: APb=22.99, APm=20.39
Epoch 7: APb=22.90, APm=20.29
Epoch 8: APb=22.31, APm=19.72
Epoch 9: APb=21.66, APm=19.14
Epoch 10: APb=21.16, APm=18.68

=====
Validation Results (best.pt)
=====
Ultralytics 8.3.39 Python-3.10.20 torch-2.12.0+cu130 CUDA:1 (NVIDIA GeForce RTX 4090, 24081MiB)
YOLOe-v8s-seg summary (fused): 238 layers, 13,660,496 parameters, 0 gradients, 40.1 GFLOPs
val: Scanning /mnt/dsc3032-gaya-shared/group7/datasets/coco_full_yolo_seg/labels/val.cache... 4952 images, 0 backgrounds, 0 corrupt: 100%|██████████| 4952/4
      Class      Images  Instances      Box(P)          R      mAP50  mAP50-95)      Mask(P)          R      mAP50  mAP50-95): 100%|██████████| 310/3

              all            4952            36335            0.542            0.204            0.309            0.218            0.538            0.202            0.303            0.193
WARNING ⚠ ConfusionMatrix plot failure: No module named 'seaborn'
WARNING ⚠ ConfusionMatrix plot failure: No module named 'seaborn'
Speed: 0.2ms preprocess, 2.2ms inference, 0.0ms loss, 1.1ms postprocess per image
Results saved to /home/gaya7/dsc3032-gaya-shared/group7/official_yoloe/runs/segment/val7
  APb  (mAP50-95) = 21.84
  APb50 (mAP50)   = 30.88
  APm  (mAP50-95) = 19.28
  APm50 (mAP50)   = 30.25
```

**Note:** Trained on the university server via tmux.  
Since tmux does not support scroll-up,  
epoch-by-epoch results are captured as screenshots.

# DEMO - Table 4

## Full Fine Tuning training log and final evaluation result

FT Full COCO (batch=16, 160 epochs) - Training Progress

| Epoch | APb   | APm   | box_loss | seg_loss |
|-------|-------|-------|----------|----------|
| 1     | 25.06 | 22.22 | 1.2028   | 2.5494   |
| 2     | 25.06 | 22.22 | 1.2043   | 2.5136   |
| 3     | 25.06 | 22.22 | 1.2099   | 2.5043   |
| 4     | 25.06 | 22.22 | 1.2029   | 2.4741   |
| 5     | 25.06 | 22.22 | 1.1914   | 2.4438   |
| 6     | 25.06 | 22.22 | 1.1801   | 2.4255   |
| 7     | 25.06 | 22.22 | 1.1710   | 2.4016   |
| 8     | 25.06 | 22.22 | 1.1646   | 2.3833   |
| 9     | 25.06 | 22.22 | 1.1580   | 2.3685   |
| 10    | 25.06 | 22.22 | 1.1506   | 2.3547   |
| 11    | 25.06 | 22.22 | 1.1478   | 2.3455   |
| 12    | 25.06 | 22.22 | 1.1402   | 2.3313   |
| 13    | 25.06 | 22.22 | 1.1386   | 2.3253   |
| 14    | 25.06 | 22.22 | 1.1337   | 2.3138   |
| 15    | 25.06 | 22.22 | 1.1303   | 2.3085   |
| 16    | 25.06 | 22.22 | 1.1292   | 2.3014   |
| 17    | 25.06 | 22.22 | 1.1261   | 2.2943   |
| 18    | 25.06 | 22.22 | 1.1242   | 2.2920   |
| 19    | 25.06 | 22.22 | 1.1213   | 2.2832   |
| 20    | 25.06 | 22.22 | 1.1184   | 2.2767   |
| 21    | 30.93 | 27.07 | 1.1173   | 2.2732   |
| 22    | 30.93 | 27.07 | 1.1171   | 2.2753   |
| 23    | 30.93 | 27.07 | 1.1155   | 2.2719   |
| 24    | 30.93 | 27.07 | 1.1133   | 2.2668   |
| 25    | 30.93 | 27.07 | 1.1120   | 2.2611   |
| 26    | 30.93 | 27.07 | 1.1103   | 2.2569   |
| 27    | 30.93 | 27.07 | 1.1099   | 2.2592   |
| 28    | 30.93 | 27.07 | 1.1067   | 2.2495   |
| 29    | 30.93 | 27.07 | 1.1075   | 2.2484   |
| 30    | 30.93 | 27.07 | 1.1041   | 2.2440   |
| 31    | 30.93 | 27.07 | 1.1041   | 2.2428   |

|    |       |       |        |        |
|----|-------|-------|--------|--------|
| 33 | 30.93 | 27.07 | 1.1005 | 2.2363 |
| 34 | 30.93 | 27.07 | 1.1005 | 2.2380 |
| 35 | 30.93 | 27.07 | 1.0982 | 2.2340 |
| 36 | 30.93 | 27.07 | 1.0990 | 2.2333 |
| 37 | 30.93 | 27.07 | 1.0984 | 2.2317 |
| 38 | 30.93 | 27.07 | 1.0979 | 2.2283 |
| 39 | 30.93 | 27.07 | 1.0941 | 2.2233 |
| 40 | 30.93 | 27.07 | 1.0956 | 2.2287 |
| 41 | 31.23 | 27.38 | 1.0954 | 2.2240 |
| 42 | 31.23 | 27.38 | 1.0944 | 2.2218 |
| 43 | 31.23 | 27.38 | 1.0934 | 2.2159 |
| 44 | 31.23 | 27.38 | 1.0932 | 2.2169 |
| 45 | 31.23 | 27.38 | 1.0911 | 2.2150 |
| 46 | 31.23 | 27.38 | 1.0922 | 2.2160 |
| 47 | 31.23 | 27.38 | 1.0895 | 2.2105 |
| 48 | 31.23 | 27.38 | 1.0895 | 2.2094 |
| 49 | 31.23 | 27.38 | 1.0864 | 2.2038 |
| 50 | 31.23 | 27.38 | 1.0878 | 2.2051 |
| 51 | 31.23 | 27.38 | 1.0868 | 2.2037 |
| 52 | 31.23 | 27.38 | 1.0853 | 2.2007 |
| 53 | 31.23 | 27.38 | 1.0848 | 2.2017 |
| 54 | 31.23 | 27.38 | 1.0843 | 2.1949 |
| 55 | 31.23 | 27.38 | 1.0831 | 2.1968 |
| 56 | 31.23 | 27.38 | 1.0830 | 2.1941 |
| 57 | 31.23 | 27.38 | 1.0824 | 2.1946 |
| 58 | 31.23 | 27.38 | 1.0799 | 2.1928 |
| 59 | 31.23 | 27.38 | 1.0808 | 2.1907 |
| 60 | 31.23 | 27.38 | 1.0817 | 2.1875 |
| 61 | 30.03 | 26.29 | 1.0798 | 2.1860 |
| 62 | 30.03 | 26.29 | 1.0793 | 2.1885 |
| 63 | 30.03 | 26.29 | 1.0778 | 2.1862 |
| 64 | 30.03 | 26.29 | 1.0773 | 2.1797 |

**Note:** Trained on the university server via tmux.  
 Since tmux does not support scroll-up,  
 epoch-by-epoch results are captured as screenshots.

# DEMO - Table 4

## Full Fine Tuning Training Log & Final Evaluation

```
=====
Validation Results (best.pt)
=====
Ultralytics 8.3.39 Python-3.10.20 torch-2.12.0+cu130 CUDA:1 (NVIDIA GeForce RTX 4090, 24081MiB)
YOLOe-v8s-seg summary (fused): 238 layers, 13,660,496 parameters, 0 gradients, 40.1 GFL0Ps
val: Scanning /mnt/dsc3032-gaya-shared/group7/datasets/coco_full_yolo_seg/labels/val.cache... 4952 images, 0 backgrounds, 0 corrupt: 100%|██████████| 4952/4
      Class      Images  Instances   Box(P          R      mAP50  mAP50-95)     Mask(P          R      mAP50  mAP50-95): 100%|██████████| 310/3
=====
              all         4952         36335         0.517         0.413         0.429         0.306         0.517         0.406         0.419         0.266
WARNING ⚠ ConfusionMatrix plot failure: No module named 'seaborn'
WARNING ⚠ ConfusionMatrix plot failure: No module named 'seaborn'
Speed: 0.2ms preprocess, 2.6ms inference, 0.0ms loss, 1.2ms postprocess per image
Results saved to /home/gaya7/dsc3032-gaya-shared/group7/official_yoloe/runs/segment/val8
APb   (mAP50-95) = 30.64
APb50 (mAP50)   = 42.90
APm   (mAP50-95) = 26.65
APm50 (mAP50)   = 41.89
(yoloe_proj) gaya7@aicoss220:~/dsc3032-gaya-shared/group7$ |
```

**Note:** We attempted 160 epochs as in the paper, but could not finish before the presentation due to compute limits. The results above are evaluated from the best checkpoint reached so far.

# CHALLENGES & SOLUTIONS

## Technical Issues

1. Annotation formats was different between YOLOE-Ultralytic and YOLO-World-MMDetection
2. Ultralytics YOLO-World wrapper produced AP 12.32, far below the paper
3. Reduced batch / epochs / dataset due to compute limits → FT underperformed
4. Broken dataset symlinks caused an empty label cache and misleading training error

## Solutions

1. Added the required file\_name field to lvis\_v1\_minival.json
2. Used official YOLO-World tools/test.py, reproducing AP 22.70
3. Scaled up (batch 64, more epochs, Full COCO) when GPU headroom available → significant AP gain
4. Verified symlinks with readlink, rebuilt them with absolute paths, and removed stale caches.

# CONCLUSION

## Paper Contributions

- RepRTA: text prompt, zero overhead
- SAVPE: visual prompts with low complexity
- LRPC: prompt-free without language model

## Our Reproduction

- Table 1 ranking reproduced
- FT > LP pattern reproduced with Full COCO
- Absolute AP gap explained by batch and epoch constraints

## Limitations

- Full FT needs larger batch (128) and more epochs (160) for full convergence
- TensorRT FPS not measured
- Evaluation settings need tighter control

**Takeaway:** The main claim is not that every number is perfectly reproduced under limited resources, but that YOLOE's efficiency-oriented design and relative advantage are visible.